

Overview of MEG/EEG analysis with MNE-Python.

EEG (Electroencephalography) → measures potential difference, resulting from ionic current flows within the neurons of the brain
 → data recovered from summation of post-synaptic potentials stored as multi-channel time-series of PD. (pV)

MEG (Magnetoencephalography) → neuroimaging tech for mapping brain activity by recording magnetic fields produced by electric currents of brain.
 → femtoTesla (fT)

• LOADING DATA

- MNE-Python format FIF other [EDF, BDF, etc].
- `mne.datasets.sample.data_path` (down datasets).
- `data_path` (checks if data local system min h ki nhi)
- ≥) `mne.io.read_raw_fit(fname, preload = false)`
 - ↓ `Path`
(data stays on disk)
 - ↓ `False`
 - ↓ `True`
(stored on RAM)
↓
used to modify the data (filtering or cleaning)

→ `read_raw_fit` shows basic info on screen.

SSP

Signal
Space
Proj

→ 'Magnetic Noise' ko cancel karne wale mathematical vectors. using matrix multiply it reduces the rank of the matrix.

→ EEG - Mean Reference → EEG signals ka ek reference point hota hai. Yeh projector uske stable ref par set karta hai.

Print (raw)

Yeh upar-upar ki info deta.

print (raw.info)

Yeh pura pipeline, saari info deta hai

Q. What does info track?

- 1) Channel Locations: Har sensor dimag pe kyun laga hai
- 2) Applied Filters: Data load karte time kounse filter lagae hai. (0-40Hz)
- 3) CHS ENTRY: MNE directly detects ki kaunsa sensor EEG hai, kaunsa MEG, kounsa EOG (Eye movement).

⇒ raw.plot()

→ Shows voltage or flux changes over time.
→ marks 'bad' channels. (Sensors that show malfunctioning or flatlines)

⇒ compute_psd() → Spectrum Analyzer.
PSD (Power Spectral density)

→ PSD plot helps to differentiate between α , β , θ freq bands. For eg α if relaxed.

* use only plot below 50Hz

• Preprocessing

→ Maxwell Filtering: For MEG - Yeh dimag ke bahar se aane wale mag noise ko block karta hai

ICA → Imagine do logo ki awaz hai same time pe ICA unhe alag alag component banata hai

`ica = mne.preprocessing.ICA(n_components=20, random_state=1, max_iter=800)`

↓
do alag parts
mein break.

`ica.fit(raw)` → MNE ne 364 se data liya aur do
- mein break kara.

→ Plot properties → Spatial Map
Spectrum
Epoch

⇒ STIM CHANNEL - The 'Input Port' for external triggers. eg 5V might mean 'Sound played' and 10V 'Button pressed'

⇒ STI 014: sum of all pulses.

32 = Button

⇒ EVENT ID: unique integer code for action (1=Audio,

⇒ EVENT DICTIONARY: a lookup table to turn integers into words.

⇒ FIXED LENGTH EVENTS: used for things that are not event-related.

`events = mne.find_events(raw, stim_channel="STI 014")`
`fig = mne.viz.plot_events()`

Epoching

↳ Dimaag continuous hota hai, par hum usse 0.7 sec chahiye jab subject ne stimulus dekha.

↳ done with a rejection dictionary.

tmin → event hone se 0.2s pehle

tmax → 0.5s bad tak.

Epochs object : (No of Epochs, No of channels, Time points)
3D Matrix banata hai

⇒ REJECT DICT : ek threshold set karke. Jaise
(agar 150 micronvolt se upar gaya toh epoch pura gaebe)

reject1 = dict(
mag = $4000e^{-15}$,
grad = $4000e^{-12}$,
eeg = $150e^{-6}$,
eog = $250e^{-6}$)

epochs = mne.epochs(
raw,
events,
event_id = event_dict,
tmin = -0.4,
tmax = 0.5,
reject = reject1,
preload = True,

→ If clean data 90 sound vs 70 light hai
to compare karunge to result biased ayega to
we equalize them.

equalize_event_counts.

⇒ plot_image

coloured map

Row → Trial epoch

Color → Amplitude

⇒ del → to delete.

TIME - FREQUENCY ANALYSIS

↳ mne.time - frequency.

frequencies = np.arange(7, 30, 3):
7Hz se lekar 30Hz tak.

average() → random noise becomes 0.

mne.viz.plot_compact-enokeds.

↳ do alag Auditory vs visual ko ek hi graph mein plot kaarti hai

Global Field Power (GFP) → Summary graph
jo ki dikhata ki overall dimag ki
energy kab peak hoti.

Plots joint: many lines dikhti hai aur dusri taraf
tum peaks pe dimag par maps dikhti hai

plot-topomap: Heatmap.

mne.combine_enokeds(weights=[1, -1])

↳ Audio-visual kaarti hai. common cancel
ho jata, only unka difference bachta.

INVERSE MODELLING

↳ used to locate where brain signals are
coming from.

→ MNE uses a linear inverse operator to
project sensor measurements to source space
→ signal/noise ratio is used for smooth
output

→ mne.minimum-norm-least-squares-inverse-operator